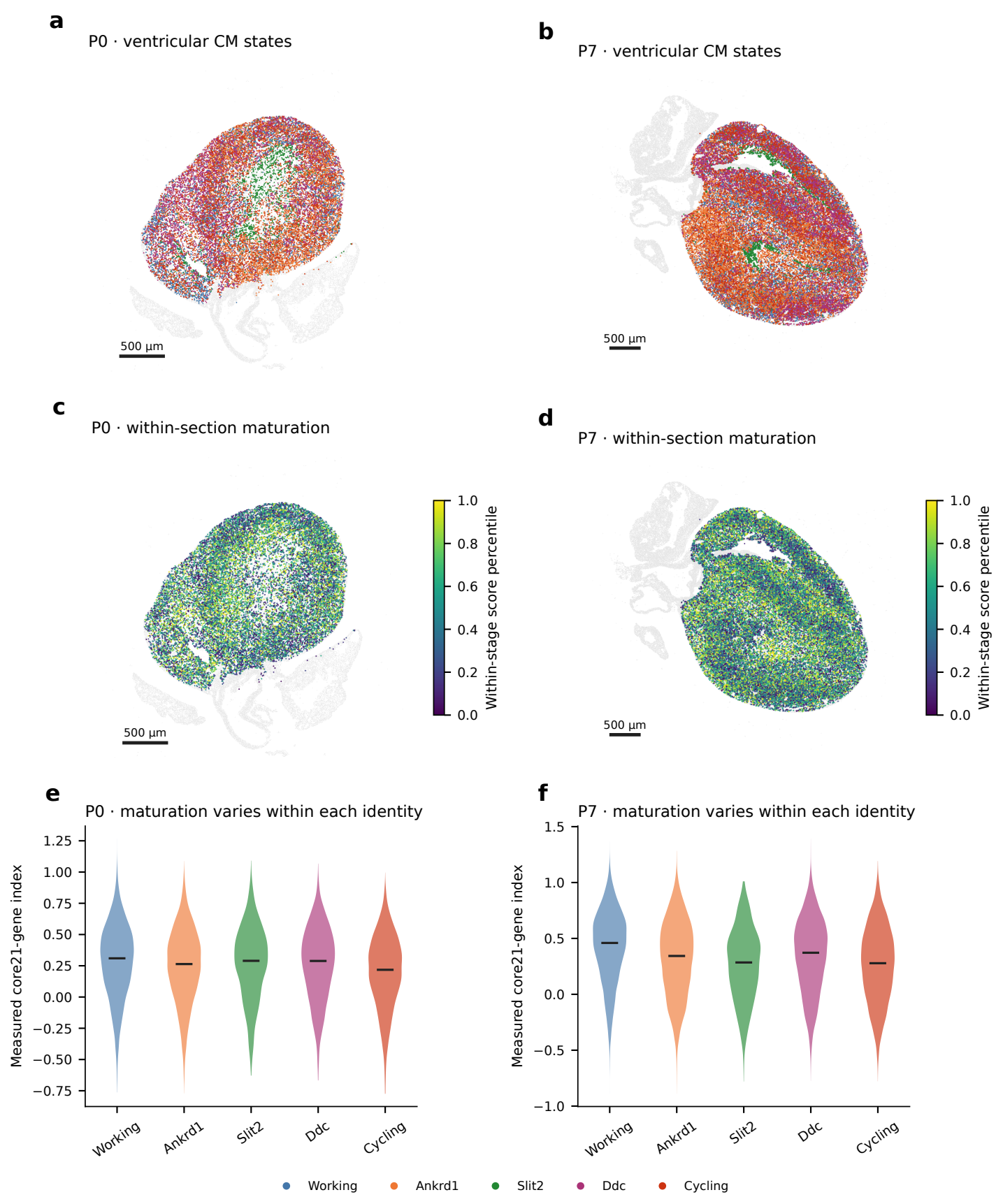
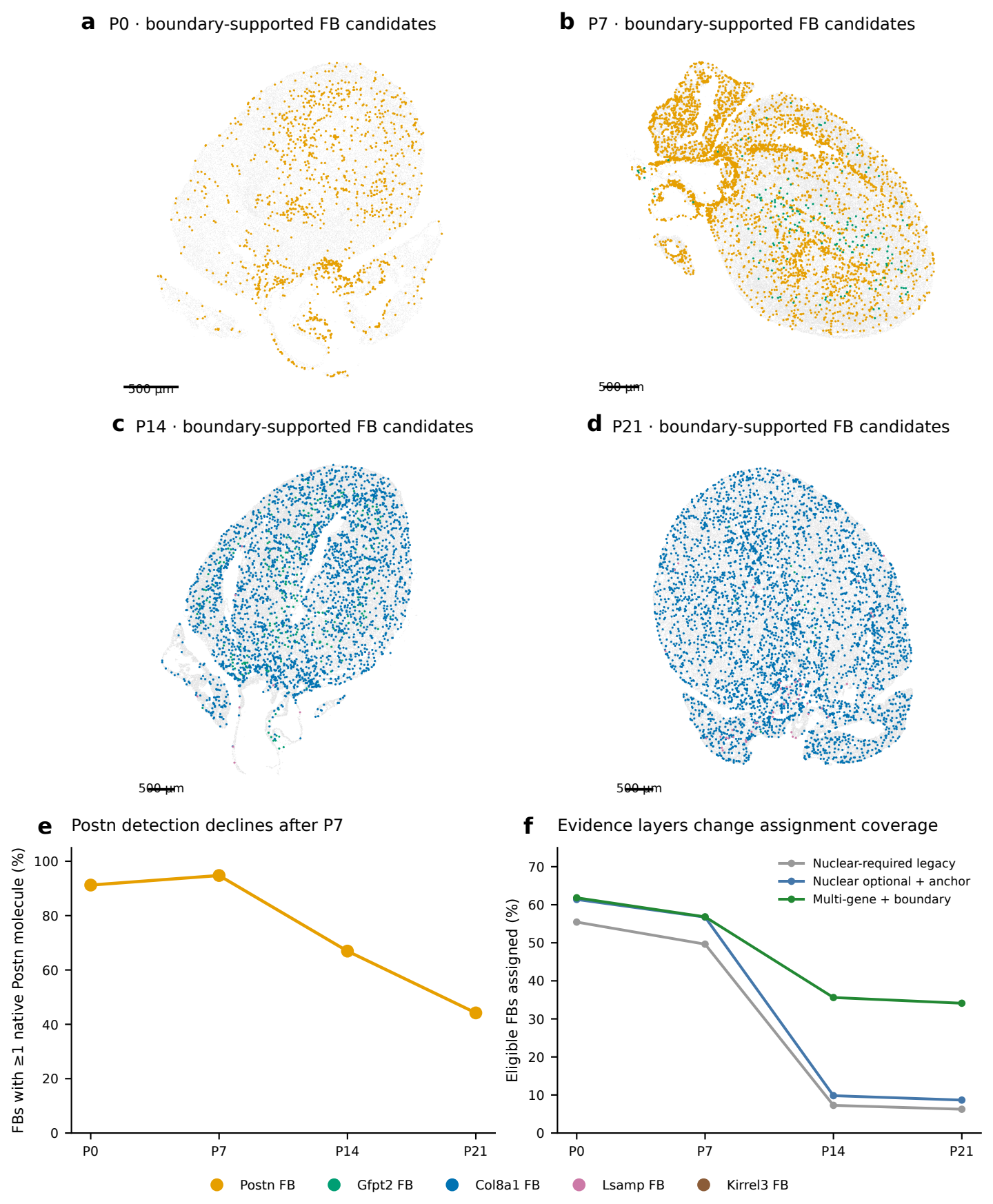
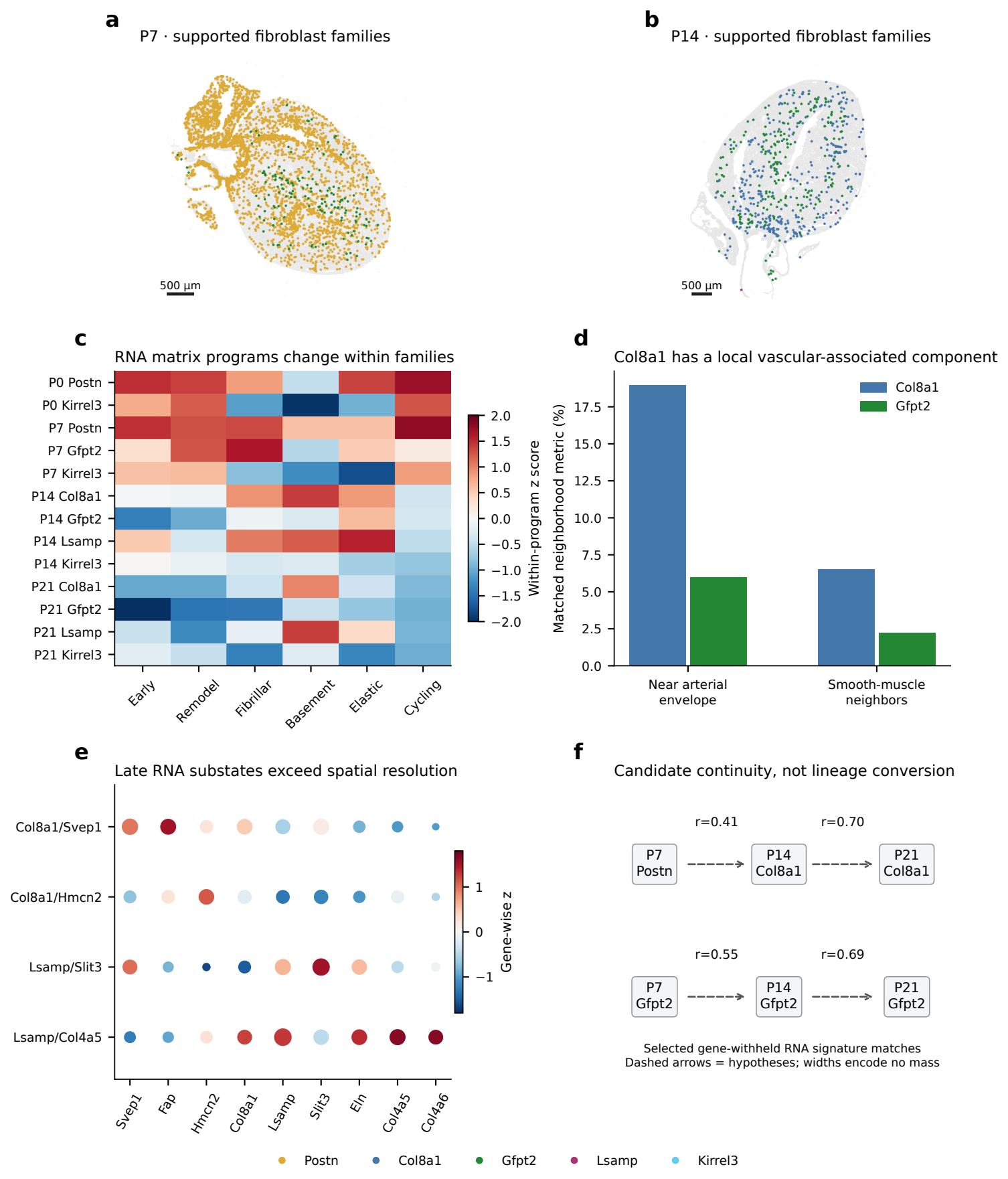
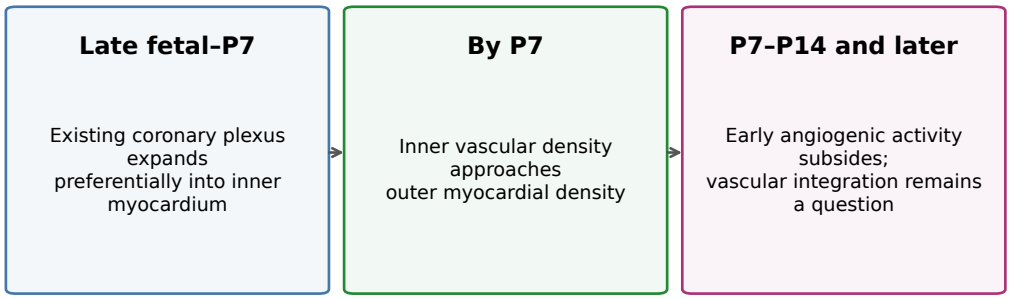

a
P0 · ventricular CM states
500 µm
b
P7 · ventricular CM states
500 µm
c
P0 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
d
P7 · within-section maturation
500 µm
1.0
0.8
0.6
0.4
0.2
0.0
Within-stage score percentile
e
P0 · maturation varies within each identity
Measured core21-gene index
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Working
Ankrd1
Slit2
Ddc
Cycling
f
P7 · maturation varies within each identity
Measured core21-gene index
1.5
1.0
0.5
0.0
−0.5
−1.0
Working
Ankrd1
Slit2
Ddc
Cycling
Working
Ankrd1
Slit2
Ddc
Cycling


a P0 · boundary-supported FB candidates
500 µm
b P7 · boundary-supported FB candidates
500 µm
c P14 · boundary-supported FB candidates
500 µm
d P21 · boundary-supported FB candidates
500 µm
e Postn detection declines after P7
FBs with ≥1 native Postn molecule (%)
0
20
40
60
80
100
P0
P7
P14
P21
f Evidence layers change assignment coverage
Eligible FBs assigned (%)
0
10
20
30
40
50
60
70
Nuclear-required legacy
Nuclear optional + anchor
Multi-gene + boundary
P0
P7
P14
P21
Postn FB
Gfpt2 FB
Col8a1 FB
Lsamp FB
Kirrel3 FB

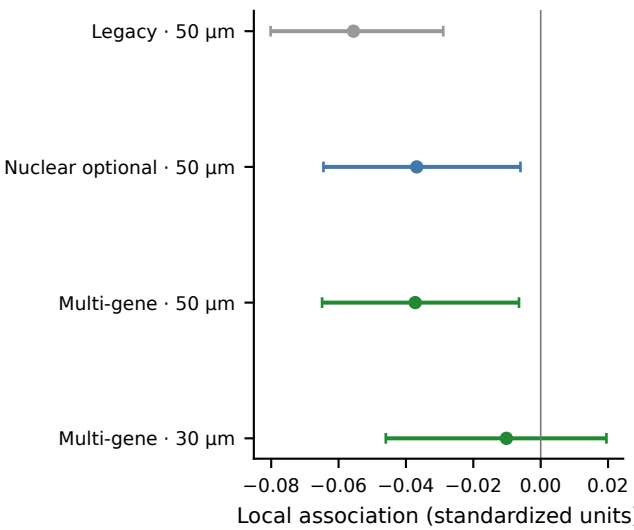

a
P7 · supported fibroblast families
500 µm
b
P14 · supported fibroblast families
500 µm
c
RNA matrix programs change within families
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early
Remodel
Fibrillar
Basement
Elastic
Cycling
Within-program z score
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
d
Col8a1 has a local vascular-associated component
Col8a1
Gfpt2
Matched neighborhood metric (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Near arterial envelope
Smooth-muscle neighbors
e
Late RNA substates exceed spatial resolution
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
Gene-wise z
1
0
−1
f
Candidate continuity, not lineage conversion
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8a1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Selected gene-withheld RNA signature matches
Dashed arrows = hypotheses; widths encode no mass
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3

a Regional vascular expansion changes the setting in which CMs develop

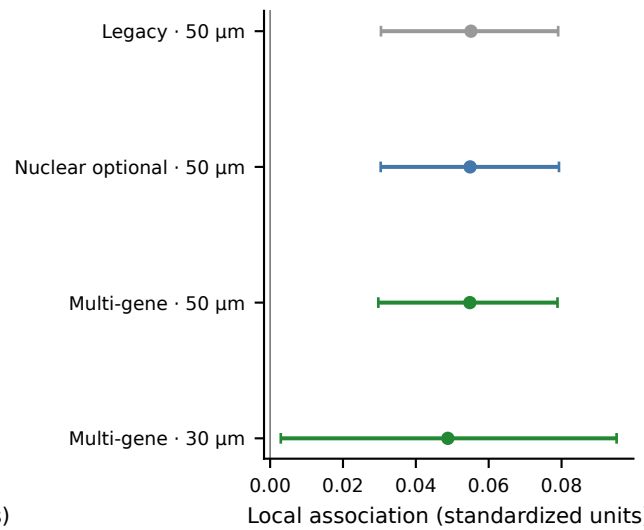


Published vascular findings (Lu et al., 2021); not newly measured perfusion in our sections

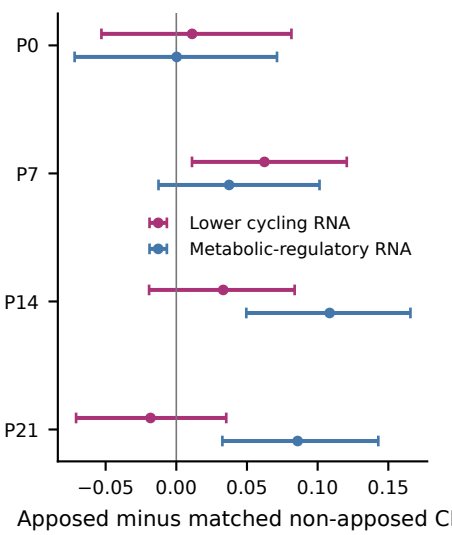
b P7 cycling CMs · Postn FB / metabolic RNA



c P14 Ankrd1 CMs · endocardial EC / sarcomere RNA



d Capillary apposition · different components over time



e Growth, access and functional integration are distinct

Growing vasculature

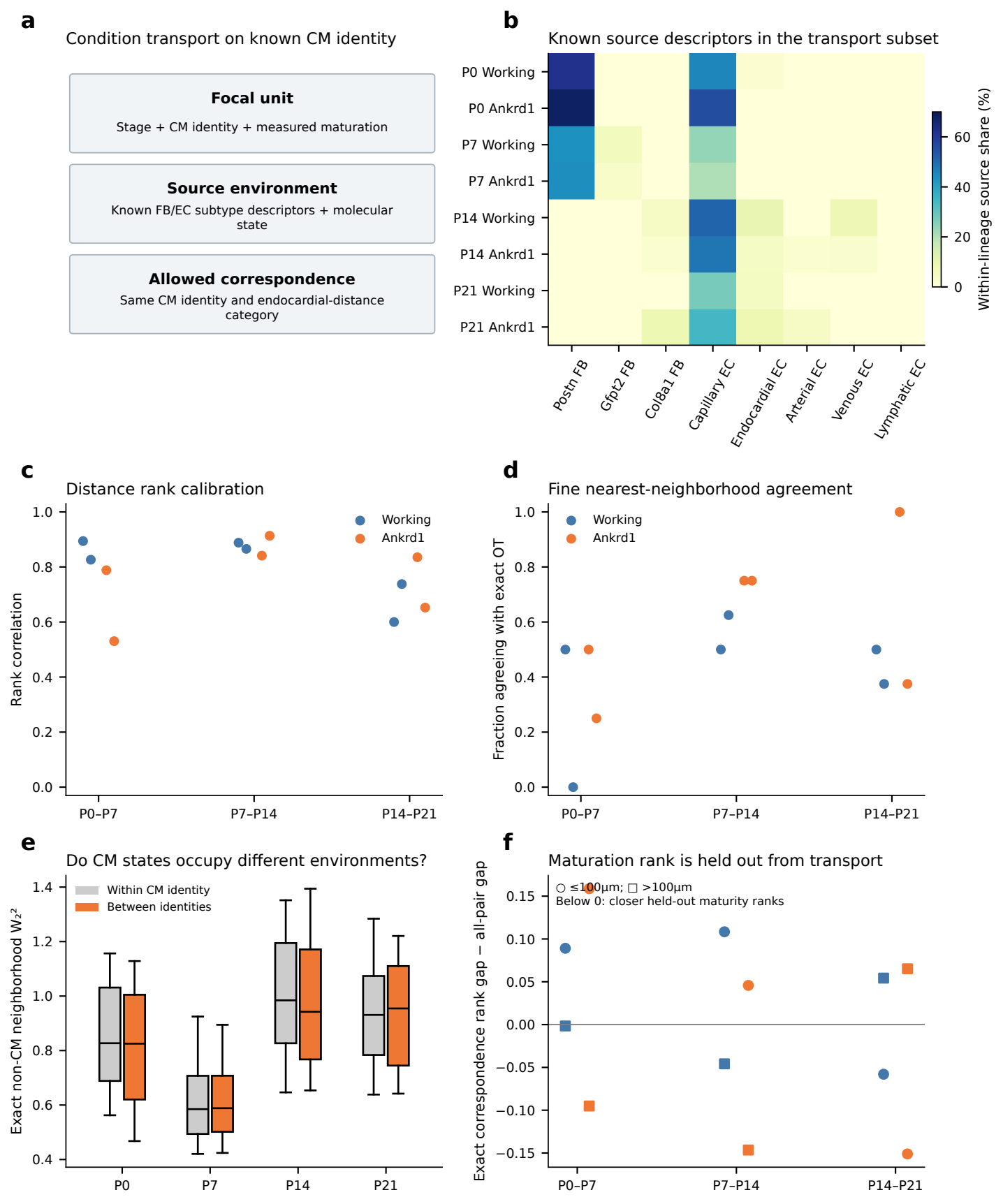
EC cycling / sprouting-associated RNA; not measured oxygen delivery

Local vascular access

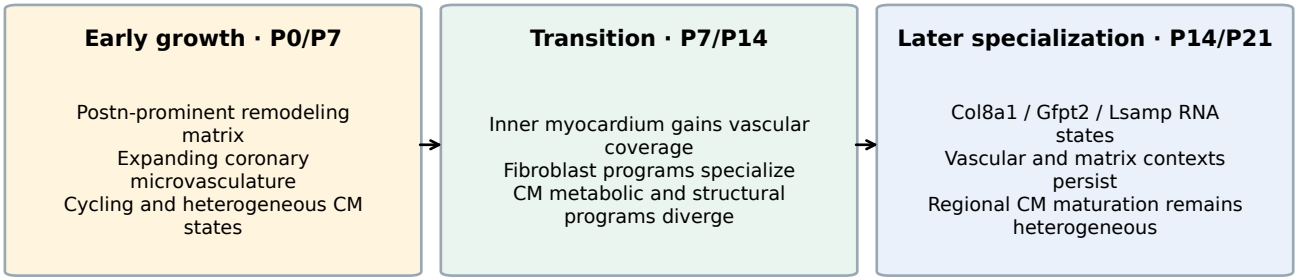
Capillary distance or boundary apposition; not proof of a perfused vessel

CM developmental response

Metabolism, sarcomeres and cycling can differ; functional maturation requires direct measurements

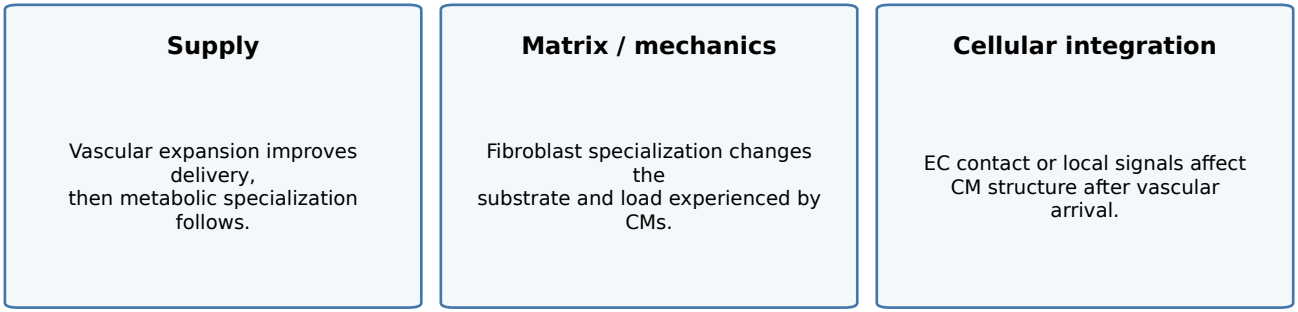

a
Condition transport on known CM identity
Focal unit
Stage + CM identity + measured maturation
Source environment
Known FB/EC subtype descriptors + molecular state
Allowed correspondence
Same CM identity and endocardial-distance category
b
Known source descriptors in the transport subset
P0 Working
P0 Ankrd1
P7 Working
P7 Ankrd1
P14 Working
P14 Ankrd1
P21 Working
P21 Ankrd1
Postn FB
Gfpt2 FB
Col8a1 FB
Capillary EC
Endocardial EC
Arterial EC
Venous EC
Lymphatic EC
Within-lineage source share (%)
0
20
40
60
c
Distance rank calibration
Working
Ankrd1
Rank correlation
P0–P7
P7–P14
P14–P21
d
Fine nearest-neighborhood agreement
Fraction agreeing with exact OT
e
Do CM states occupy different environments?
Within CM identity
Between identities
Exact non-CM neighborhood W₂²
P0
P7
P14
P21
f
Maturation rank is held out from transport
○ ≤100µm; □ >100µm
Below 0: closer held-out maturity ranks
Exact correspondence rank gap − all-pair gap

a A developmental model: coordinated changes with different local timing



Proposed synthesis; arrows denote developmental windows, not observed cell conversion or a causal chain

b Three competing explanations for regional CM development



Ankrd1 and working CMs may respond differently within the same anatomical region

c Experiments that distinguish these explanations

P3/P5/P10 in independent hearts

Co-measure perfusion, hypoxia, CM state and matrix organization; resolve temporal order.

Match the starting CM identity

Separate perfusion/oxygen, EC contact and fibroblast-derived matrix; measure respiration, calcium and force.

Trace proposed state continuity

Test early-to-late fibroblast ancestry; use transport to nominate comparisons, not to establish fate.